

Steps I Followed to Build the Circuits in Proteus

1. How I Placed the Resistors

- First, I clicked on **Component Mode** (the yellow triangle symbol on the left toolbar).
- Then, I clicked the '**P**' button to open the Pick Devices window.
- In the search box, I typed **RES** to find the standard resistor.
- I double-clicked it to add it to my list, and then I clicked on the workspace to place it in my circuit.
- To change the resistance values (like 3k, 5k, or 10k), I double-clicked each resistor and typed the required number in the **Resistance** box.

2. How I Added the DC Power Source (Battery)

- I went back to **Component Mode** and clicked the '**P**' button again.
- I typed **BATTERY** in the search box.
- I double-clicked the battery symbol to add it and placed it on my workspace.
- To set the correct voltage, I double-clicked the battery and updated the **Voltage** box to 9V.

3. How I Added the AC Power Source (VSINE)

- While in **Component Mode**, I clicked the '**P**' button.
- I typed **VSINE** in the search box to find the AC sine wave generator.
- After placing it on the workspace, I double-clicked it to open its properties.
- To get an exact 5V output on the meters, I set the **Amplitude** to 7.07 and the **Frequency** to 50.
- I also changed the **Part Value** box to "5V AC" to make the schematic easy to read.

4. How I Placed the Measuring Meters

- I went to the left toolbar and clicked on **Instruments Mode** (the icon that looks like a small multimeter).
- A list appeared, and I selected the meters I needed for the specific task (**DC Voltmeter**, **AC Voltmeter**, **DC Ammeter**, or **AC Ammeter**).
- I clicked on the workspace to place them and wired them into the circuit.

- **Important:** For all the ammeters, I double-clicked them and changed the **Display Range** to **Milliamps** so they would show accurate readings instead of just 0.00.

5. How I Added the Ground Connection

- Finally, I clicked on **Terminals Mode** on the left toolbar (the icon with two yellow arrows pointing opposite ways).
- I selected **GROUND** from the list.
- I placed it at the bottom of my circuit and wired it to the negative return path of my power source to complete the setup.

Screenshots of the compartments are attached below:



